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Configuring Keycloak in Pulse with OpenID Connect

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This page describes how to integrate Keycloak as the Identity Provider (IdP) into the Pulse environments using OpenID Connect. The integration has been tested for Keycloak version 20.0.1. Take this into consideration when following this guide and the official Keycloak documentation.

Connecting Feedzai products to Keycloak

Pulse real-time scoring

Keycloak Configuration

To configure Keycloak to accept Pulse connections, complete the following steps:

1. Create a new OpenID Connect client, filling a descriptive Client ID (from now on referred as `CLIENT_ID`), e.g. `pulse-oidc`, and optionally define Name and Description. Before saving, and referring to the HTTPS URL of the real-time scoring Pulse environment as `PULSE_SCORING_URL` , do the following changes to the default configuration:
 1. Enable Client authentication (sets this client as *confidential*) and Service accounts roles.
 2. Set Root URL to `{PULSE_SCORING_URL}` .
 3. Set Valid Redirect URIs to `{PULSE_SCORING_URL}/*` .
2. Save the client and go to the Client scopes tab to configure the claims to include in JWTs besides the standard ones.
3. Create a new OpenID Connect Client scope (or change an existing one) such that the following claims are included:

JWT Claim	Mapper	Required?	Where to add
preferred_username	username (predefined/built-in)	Yes	Access token
aud	audience resolve (predefined/built-in)	Required if the server validating the tokens requires audience claim	
name	full name (predefined/built-in)	No, but recommended, so that the user's Full Name is available to Pulse	
email	email (predefined/built-in)	Required when using features that send emails to users (e.g. 4eye check); otherwise, recommended	
locale	locale (predefined/built-in)	No, but highly recommended if internationalization is enabled at the IdP and the locale is used in Pulse	
<code>roles.\$CLIENT_ID</code> is the default in Pulse, but can be configured	client roles (predefined/built-in)	Yes, because the users need roles to be able to access the Pulse UI	Userinfo
		Required only when users are associated with groups and	

JWT Claim	Mapper	Required?	Where to add
<code>groups</code> is the default in Pulse, but can be configured	Custom mapper created by configuration with Mapper Type "Group Membership"; "Full group path" must be disabled	that information is to be propagated to Pulse	
<code>tenants</code> is the default in Pulse, but can be configured	Having tenants associated with users is not a concept of Keycloak; the workaround is to have that information as a user attribute, and then use a custom mapper created by configuration with Mapper Type "User Property", to map the value of that attribute to the "tenants" claim	Required only when users are associated with tenants and that information is to be propagated to Pulse	

4. Save the Client scope and add it to the client for Pulse, making sure it is set as "Default".
5. Configure other settings, such as token/session durations, flows, etc.
6. Save the client.

Pulse properties

Create the following properties to configure Pulse to connect to Keycloak for SSO using OpenID Connect.

The client secret (referred as `CLIENT_SECRET` in the configuration example below) can be retrieved at the Credentials tab of the client in Keycloak. Since this is sensitive information, the corresponding property in Pulse has encryption enabled. Therefore, if Pulse has an encryption key for properties set up, then the value of this property needs to be encrypted using that key.

The `discoveryURI` (referred as `DISCOVERY_URI` in the configuration example below) is the URL of the "well known" endpoint of the IdP, and appears at the Realm Settings page of Keycloak as a link to "OpenID Endpoint Configuration".

```
#####
#                               OIDC
#####
pulse.global.oidc.clientId=<CLIENT_ID>
pulse.global.oidc.secret=<CLIENT_SECRET>
pulse.global.oidc.discoveryURI=<DISCOVERY_URI>

#####
#                               JWT Auth Provider
#####
# The following property is the id of the configuration, only needed to enable the plugin
# if no other property is set
pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.id=feedzai-keycloak

# Defaults for connection timeouts to the IdP and JWT clock skew
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.connectTimeoutMs=500;
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.readTimeoutMs=500;
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.maxClockSkewSeconds=60;

# Default values for JWT claim names/paths, for clientId = `CLIENT_ID`
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.tenantsPath=/tenants
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.groupsPath=/groups
#pulse.manager.services.core.security.authplugin.provider.jwt.idpConfig.rolesPath=/roles/
<CLIENT_ID>
```

